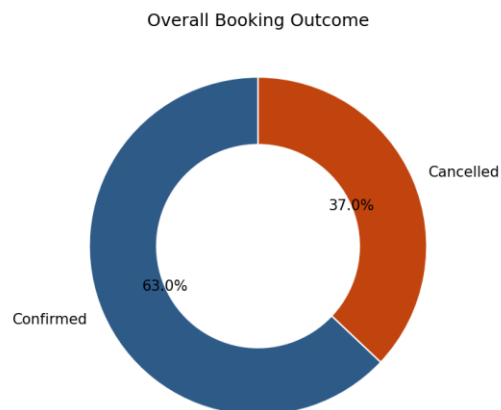


HOTEL BOOKING ANALYSIS & CANCELLATION INSIGHTS

An End-to-End Analytics Project using Python, SQL & Power BI



Dataset: 119,390 Hotel Bookings | 32 Attributes
Dashboards: Hotel Booking Analysis & Cancellation Analysis

Prepared July 2026

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1. Project Overview

This project delivers an end-to-end analysis of hotel booking behavior, combining Python for data preparation and exploration, SQL for structured business querying, and Power BI for interactive visualization. The objective is to understand what drives booking cancellations, quantify their financial and operational impact, and translate the findings into concrete, actionable recommendations for hotel management.

Two dedicated Power BI dashboards were built as part of this project:

- Hotel Booking Analysis Dashboard — an operational overview of booking volume, guest behavior, and revenue-adjacent KPIs.
- Cancellation Analysis Dashboard — a focused, drill-down view of cancellation drivers across hotel type, market segment, deposit policy, customer history, and booking lifecycle attributes.

Together, these dashboards give both a high-level pulse on performance and a diagnostic lens into why guests cancel, supporting data-driven decisions across revenue management, front-office operations, and marketing.

1.1 Objectives

- Quantify overall booking volume, confirmation rate, and cancellation rate.
- Identify the segments, channels, and policies most associated with cancellations.
- Understand how lead time, prior guest history, and engagement signals (special requests, booking changes) relate to cancellation likelihood.
- Provide the business with a prioritized set of recommendations to reduce cancellations and protect revenue.

1.2 Tools & Technologies

Layer	Tool	Purpose
Data preparation & EDA	Python (pandas, numpy, matplotlib)	Cleaning, feature engineering, exploratory analysis
Business querying	SQL	Aggregation, KPI calculation, segment comparisons
Visualization	Power BI	Interactive dashboards for stakeholders

2. Dataset Summary

The analysis is based on a hotel booking dataset containing 119,390 records across 32 attributes describing hotel type, booking channel, guest profile, pricing, and cancellation status. The working extract used for this analytics pipeline focuses on the core operational and behavioral fields listed below, which anchor the SQL business analysis and Power BI dashboards.

2.1 Key Fields

Field	Description
Hotel	Hotel type — City Hotel or Resort Hotel
Lead Time	Days between booking date and arrival date
Market Segment	Booking channel/segment (Online TA, Groups, Direct, Corporate, etc.)
Deposit Type	No Deposit, Non Refund, or Refundable
ADR	Average Daily Rate — average revenue per occupied room-night
Customer Type	Contract, Group, Transient, Transient-Party
Previous Cancellations	Number of prior bookings the guest cancelled
Previous Bookings Not Canceled	Number of prior bookings the guest honored
Booking Changes	Number of amendments made to the booking
Days in Waiting List	Days a booking spent on the waiting list before confirmation
Total of Special Requests	Count of special requests made by the guest
is_canceled	Target flag — 1 if the booking was cancelled, 0 if honored

2.2 Dataset at a Glance

119,390 Total Bookings	75,166 Confirmed	44,224 Cancelled	37.0% Cancellation Rate
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Roughly 1 in every 2.7 bookings in the dataset ends in cancellation. This is materially higher than typical industry benchmarks (commonly cited in the 20-25% range) and underlines why a structured cancellation analysis is a priority for revenue protection.

3. Exploratory Data Analysis using Python

Exploratory Data Analysis (EDA) was carried out in Python using pandas for data manipulation and matplotlib for visual checks. The goal at this stage was to make the dataset analysis-ready and to surface early patterns before moving into SQL-based aggregation and Power BI visualization.

3.1 Data Preparation Steps

- Loaded the raw booking records into a pandas DataFrame and inspected shape, data types, and value ranges.
- Checked for and handled missing values across all fields to ensure downstream aggregations were not distorted.
- Identified and removed duplicate booking records to avoid double-counting in KPIs.
- Standardized column names and categorical labels (e.g., consistent casing and naming for hotel type, market segment, and deposit type) for consistent grouping.
- Engineered a Lead Time Group feature by binning the continuous lead_time field into five buckets: 0-30, 31-90, 91-180, 181-365, and 365+ days, enabling trend analysis without noisy day-by-day granularity.
- Engineered supporting behavioral flags — such as whether a guest had any prior cancellations, whether a booking spent time on the waiting list, and whether any booking changes or special requests were made — to simplify comparative analysis.
- Exported the cleaned, feature-enriched dataset for consumption by SQL scripts and Power BI.

3.2 Lead Time Distribution

Lead time — how far in advance a guest books — proved to be one of the strongest behavioral signals in the dataset. The chart below shows the cancellation rate rising steadily as lead time increases.

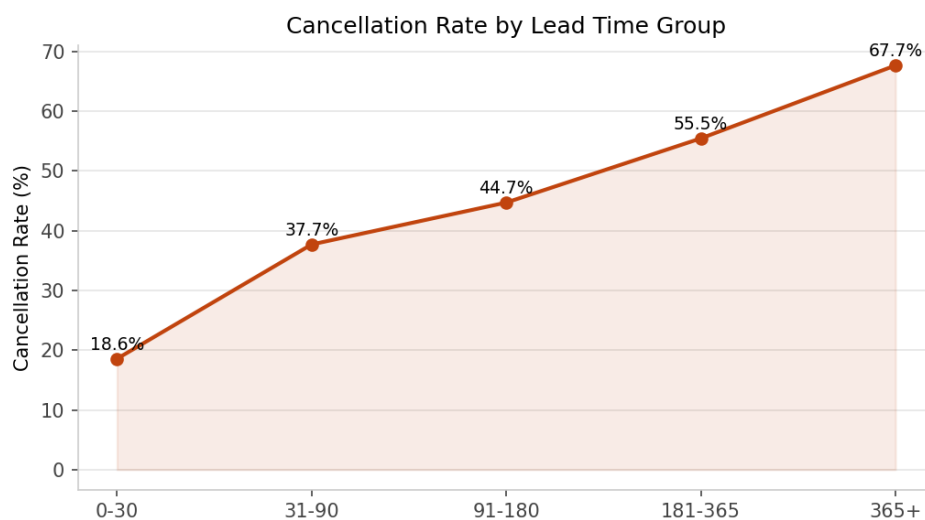


Figure 3.1 — Cancellation rate increases sharply with booking lead time.

Bookings made 365+ days in advance cancel at more than 3.5x the rate of last-minute bookings (0-30 days), suggesting that guests who book far ahead have more time and opportunity to change plans.

4. Data Analysis using SQL

With a clean dataset in place, SQL was used to answer specific business questions through aggregation and grouping. This mirrors how a BI or analytics team would query a production data warehouse to generate the KPIs feeding a dashboard.

4.1 Core Business Metrics

The queries below establish the headline KPIs used throughout the dashboards.

```
-- Overall booking KPIs
SELECT
    COUNT(*)                AS total_bookings,
    SUM(is_canceled)         AS cancelled_bookings,
    COUNT(*) - SUM(is_canceled) AS confirmed_bookings,
    ROUND(SUM(is_canceled) * 100.0 / COUNT(*), 2) AS cancellation_rate_pct,
    ROUND(AVG(lead_time), 2) AS avg_lead_time,
    ROUND(AVG(total_of_special_requests), 2) AS avg_special_requests
FROM bookings;
```

Result

total_bookings	cancelled_bookings	confirmed_bookings	cancellation_rate_pct	avg_lead_time
119,390	44,224	75,166	37.04	104.01

4.2 Bookings and Cancellations by Hotel Type

```
SELECT hotel,
    COUNT(*)
    AS bookings,
    SUM(is_canceled)
    AS cancelled,

    ROUND(SUM(is_canceled) *
    100.0 / COUNT(*),2) AS
    cancel_rate_pct
FROM bookings
GROUP BY hotel
ORDER BY cancel_rate_pct
DESC;
```

Hotel	Bookings	Cancelled	Cancel Rate
City Hotel	79,330	33,102	41.73%
Resort Hotel	40,060	11,122	27.76%

City Hotel bookings cancel at nearly 42%, roughly 14 percentage points higher than Resort Hotel. City-hotel demand skews toward business and transient travel, which tends to be more schedule-sensitive and price-shopped, both of which raise cancellation risk.

4.3 Cancellations by Market Segment

```
SELECT market_segment,
       COUNT(*) AS
bookings,

SUM(is_canceled) AS
cancelled,

ROUND(SUM(is_canceled)
* 100.0 / COUNT(*),2)
AS cancel_rate_pct
FROM bookings
GROUP BY
market_segment
ORDER BY bookings
DESC;
```

Market Segment	Bookings	Cancelled	Cancel Rate
Online TA	56,477	20,739	36.72%
Offline TA/TO	24,219	8,311	34.32%
Groups	19,811	12,097	61.06%
Direct	12,606	1,934	15.34%
Corporate	5,295	992	18.73%
Aviation	237	52	21.94%
Complementary	743	97	13.06%

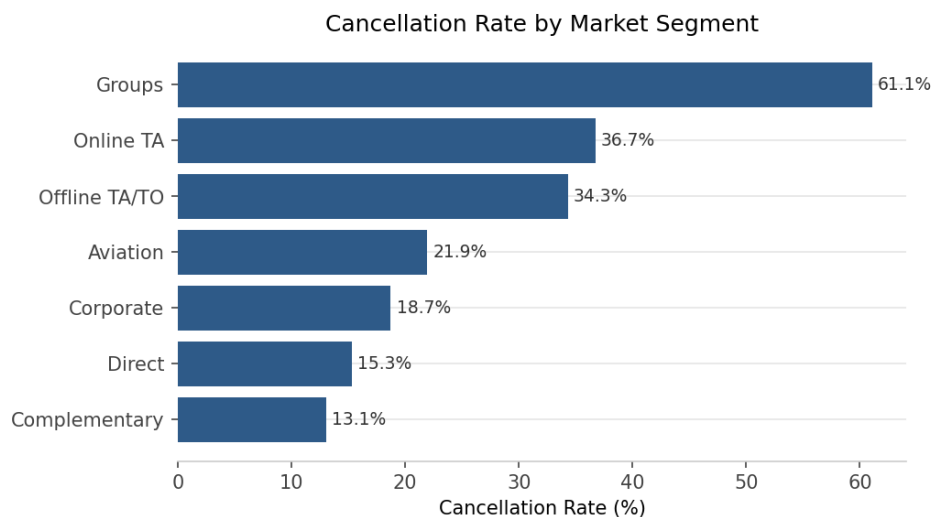


Figure 4.1 — Cancellation rate by market segment. Groups cancel at more than 4x the rate of Direct bookings.

Groups is both a high-volume and high-risk segment, cancelling at 61%, while Direct and Complementary bookings are the most reliable channels. This gap points to tentative block-holding behavior in group bookings that is rarely seen in direct guest relationships.

4.4 Cancellations by Deposit Type

```

SELECT deposit_type,
       COUNT(*) AS
bookings,

SUM(is_canceled) AS
cancelled,

ROUND(SUM(is_canceled)
* 100.0 / COUNT(*),2)
AS cancel_rate_pct
FROM bookings
GROUP BY deposit_type
ORDER BY
cancel_rate_pct DESC;

```

Deposit Type	Bookings	Cancelled	Cancel Rate
Non Refund	14,587	14,494	99.36%
Refundable	162	36	22.22%
No Deposit	104,641	29,694	28.38%

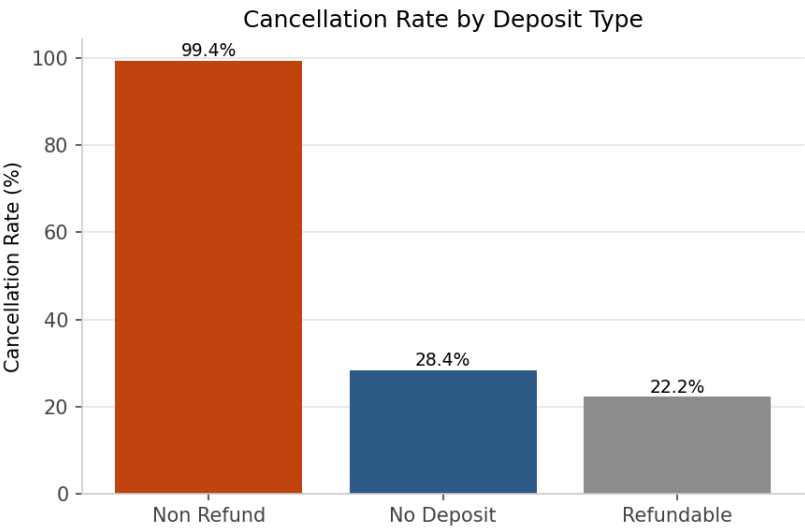


Figure 4.2 — Non-refundable bookings cancel at a near-total 99.36% rate.

This is the single most striking finding in the dataset: Non Refund bookings cancel almost universally (99.4%), the opposite of what the label implies. In practice this pattern is strongly associated with speculative or fraudulent block bookings rather than genuine guest intent, and warrants targeted policy and fraud-control review rather than being treated as a normal guest-behavior segment.

4.5 Monthly and Lead-Time Trends

With date-level granularity, monthly booking and cancellation trends can be tracked to identify seasonal peaks in demand and cancellation risk (visualized in the Power BI dashboards). Within the current extract, lead-time trend is used as the primary time-based driver:

Lead Time Group	Bookings	Cancelled	Cancel Rate
0-30 days	38,706	7,185	18.56%
31-90 days	29,553	11,141	37.70%
91-180 days	26,439	11,821	44.71%
181-365 days	21,544	11,947	55.45%
365+ days	3,148	2,130	67.66%

4.6 Previous Cancellations & Guest History

```
SELECT CASE WHEN previous_cancellations > 0
      THEN 'Had Prior Cancellations' ELSE 'No Prior Cancellations' END AS
guest_history,
      COUNT(*) AS bookings,
      SUM(is_canceled) AS cancelled,
      ROUND(SUM(is_canceled) * 100.0 / COUNT(*),2) AS cancel_rate_pct
FROM bookings
GROUP BY guest_history;
```

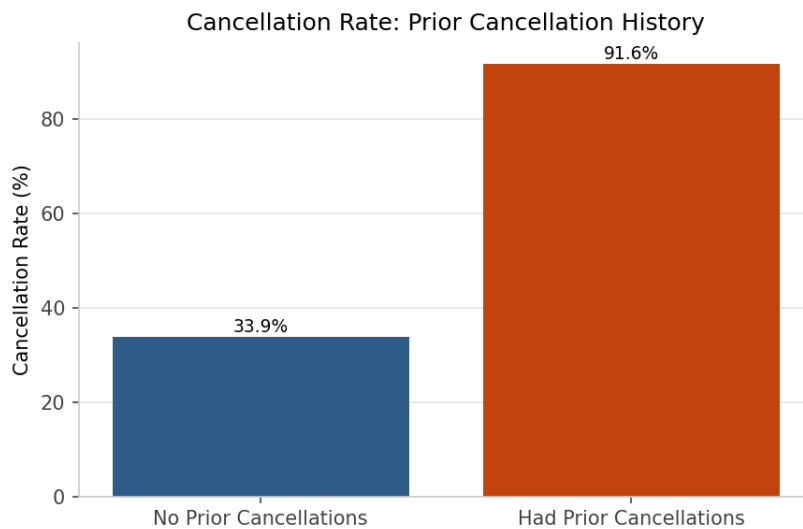


Figure 4.3 — Guests with a history of cancelling do so again at 91.6%, vs 33.9% for guests with no such history.

Cancellation history is the strongest individual predictor in the dataset: guests who have cancelled before are more than 2.5x as likely to cancel again. This makes prior-cancellation count a high-value flag for proactive retention or stricter deposit terms at time of booking.

4.7 Waiting List Analysis

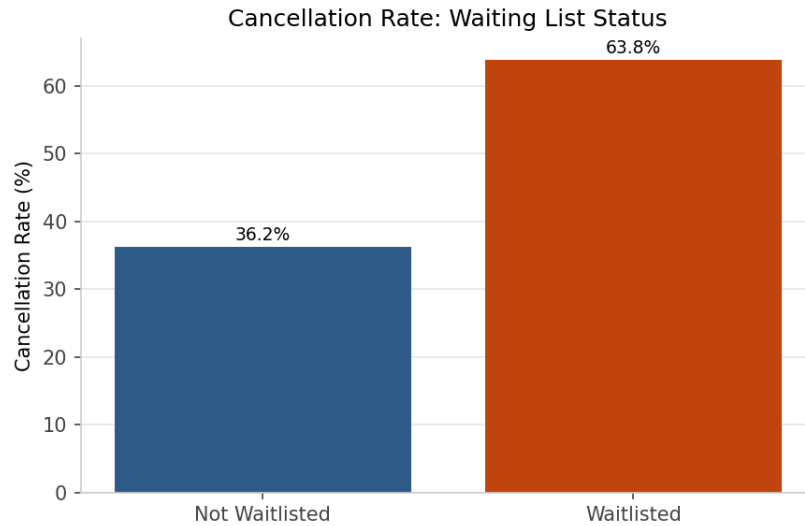


Figure 4.4 — Waitlisted bookings cancel at 63.8% vs 36.2% for bookings confirmed immediately.

Bookings that spend time on the waiting list (averaging roughly 75 days in queue before confirmation) cancel at nearly double the rate of instantly-confirmed bookings. The uncertainty of a pending confirmation appears to erode guest commitment the longer it persists.

4.8 Booking Changes & Special Requests

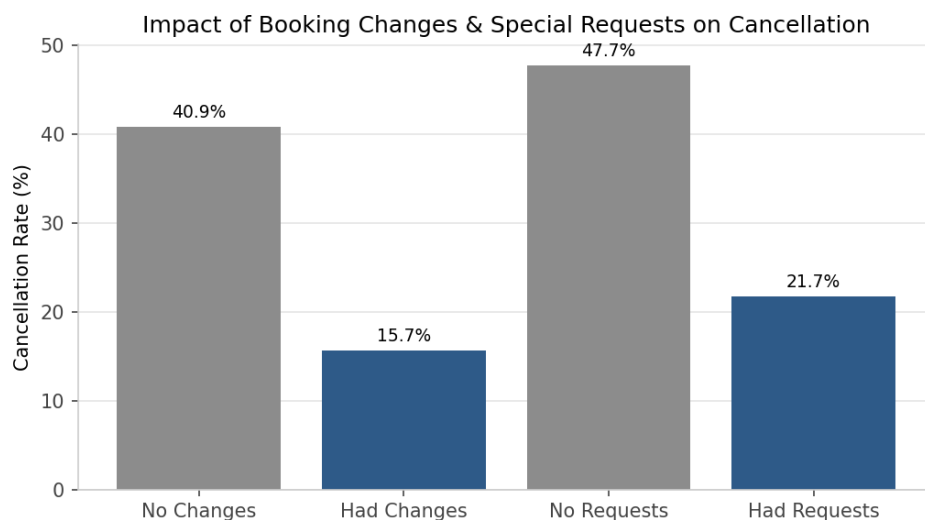


Figure 4.5 — Booking amendments and special requests are both associated with lower cancellation.

Both signals point the same direction: guests who actively engage with their booking — by amending details or adding special requests — are substantially less likely to cancel. Bookings with at least one amendment cancel at 15.7% versus 40.9% for untouched bookings, and bookings with a special request cancel at 21.7% versus 47.7% without one. This engagement pattern is a useful early-warning (or reassurance) signal for front-desk and CRM teams.

5. Dashboard in Power BI

The SQL-derived KPIs and Python-engineered features feed two purpose-built Power BI dashboards (delivered as .pbit template files): the Hotel Booking Analysis Dashboard for operational monitoring and the Cancellation Analysis Dashboard for cancellation diagnostics. Both are fully interactive, with slicers for hotel type, market segment, deposit type, and lead time group.

5.1 Dashboard 1 — Hotel Booking Analysis

This dashboard gives stakeholders a single-page operational overview of booking volume and guest behavior.

- KPI Cards: Total Bookings, Cancellation Rate, Average ADR, Average Lead Time, Average Special Requests.
- Bookings by Hotel — City Hotel vs Resort Hotel volume comparison.
- Bookings by Market Segment — channel mix across Online TA, Offline TA/TO, Groups, Direct, Corporate, and others.
- Bookings by Deposit Type — distribution across No Deposit, Non Refund, and Refundable policies.
- Monthly Trends — booking volume seasonality across the calendar year.
- Bookings by Customer Type — Transient, Transient-Party, Contract, and Group guest mix.
- Lead Time Distribution — spread of booking lead times across the five defined buckets.

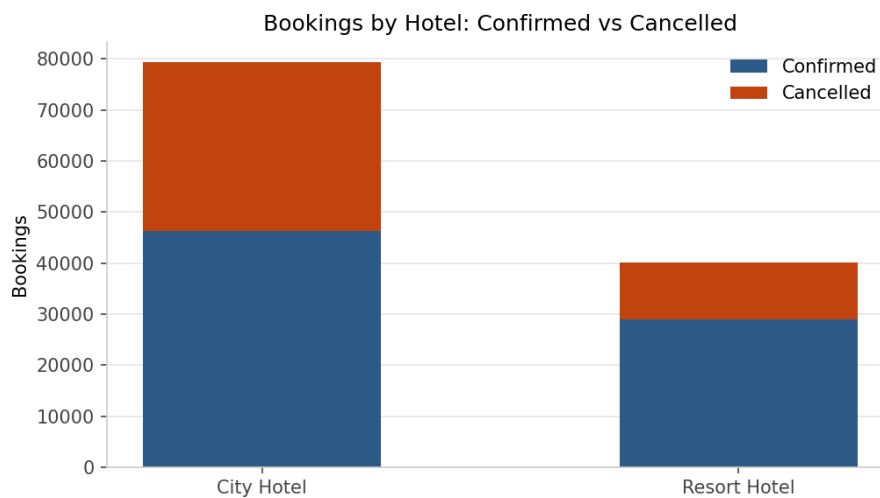


Figure 5.1 — Booking volume by hotel type, split by confirmed vs cancelled (as visualized on Dashboard 1).

5.2 Dashboard 2 — Cancellation Analysis

This dashboard drills into the drivers of cancellation identified in Section 4, letting analysts filter and compare cancellation rate across every major dimension.

- Cancelled vs Confirmed Bookings — headline split with drill-through to segment detail.
- Cancellation by Hotel — City vs Resort cancellation rate comparison.
- Cancellation by Market Segment — channel-level risk ranking.

- Cancellation by Deposit Type — policy-level risk ranking, highlighting the Non Refund anomaly.
- Cancellation by Customer Type — guest-profile risk comparison.
- Cancellation by Lead Time Group — trend of rising risk with longer booking horizons.
- Previous Cancellations — repeat-cancellation risk by guest history.
- Booking Changes — engagement vs cancellation relationship.
- Waiting List Analysis — confirmation-delay impact on cancellation.

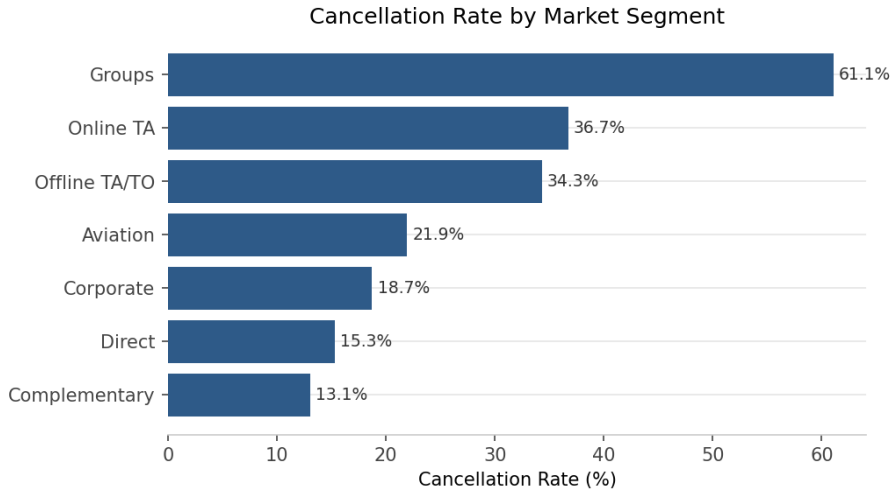


Figure 5.2 — Market segment cancellation ranking, a core visual on Dashboard 2.

5.3 Design Notes

Both dashboards use consistent color coding (blue for confirmed/stable metrics, red-orange for cancellation/risk metrics) so that risk signals are visually distinct at a glance. Cross-filtering is enabled across all visuals, so selecting a hotel type, segment, or lead-time bucket on any chart filters every other visual on the page, allowing rapid root-cause drill-down.

6. Business Recommendations

Based on the patterns surfaced across the Python EDA, SQL analysis, and Power BI dashboards, the following actions are recommended to reduce cancellations and protect revenue.

6.1 Reduce Long Lead-Time Cancellations

Cancellation rate climbs from 18.6% for bookings within 30 days to 67.7% for bookings made a year or more in advance. Introduce a staged confirmation process for long-lead bookings — e.g., a courtesy reconfirmation email or a small non-punitive deposit at the 90- and 30-day marks — to keep distant bookings engaged and reduce silent drop-off.

6.2 Improve Deposit Policies

The near-total 99.4% cancellation rate on Non Refund bookings is anomalous and should be investigated as a potential channel-quality or fraud issue rather than accepted as guest behavior. Recommend auditing the source of Non Refund bookings (likely specific OTA/group channels), tightening validation on these bookings, and considering a moderate, clearly-communicated deposit tier for No Deposit bookings in high-risk segments instead of an all-or-nothing model.

6.3 Reward Loyal, Reliable Customers

Guests with no prior cancellation history cancel at 33.9% versus 91.6% for repeat cancellers. Implement a light-touch loyalty or reliability program — priority upgrades, waived minor change fees — for guests with a track record of honoring bookings, while applying stricter deposit or confirmation requirements to guests with two or more historical cancellations.

6.4 Target High-Cancellation Segments

Groups cancel at 61.1%, more than 4x the Direct channel's 15.3%. Work with group/travel-agency partners on firmer block-release timelines and partial-deposit terms for group bookings, and shift marketing investment toward Direct and Corporate channels, which combine strong volume with the lowest cancellation risk.

6.5 Optimize Pricing and Waitlist Management

Waitlisted bookings cancel at 63.8%, nearly double the 36.2% rate for instantly confirmed bookings, and the average wait is about 75 days. Prioritize faster confirmation turnaround (via dynamic inventory management or overbooking buffers) to shorten time on the waiting list, which should directly reduce the drop-off associated with prolonged uncertainty.

6.6 Personalize the Guest Experience

Bookings with at least one special request or amendment cancel at roughly half the rate of untouched bookings. Encourage early engagement — prompting guests to add preferences or requests at time of booking — through a simple post-booking email nudge, using this behavior both

as a light retention lever and as an early signal of booking intent for front-desk and CRM prioritization.

7. Conclusion

This project demonstrates a complete analytics workflow — from raw data to business decision — spanning Python for preparation and exploration, SQL for structured business analysis, and Power BI for interactive visualization. Across 119,390 bookings, the analysis surfaced a 37.0% overall cancellation rate driven by a consistent set of factors: long lead times, group and OTA-heavy channels, non-refundable deposit terms, absence of prior guest history, and low guest engagement at time of booking.

The two Power BI dashboards operationalize these findings — the Hotel Booking Analysis dashboard for day-to-day monitoring of volume and guest mix, and the Cancellation Analysis dashboard for targeted diagnosis of cancellation drivers. Together with the recommendations in Section 6, they give hotel management a repeatable, data-driven foundation for improving occupancy, reducing revenue leakage from cancellations, and delivering a more personalized guest experience.

Prepared as part of the Hotel Booking Analysis & Cancellation Insights project — Python · SQL · Power BI.